

Make it till you fake it: Construction-centric computational framework for simultaneous image synthetization and multimodal labeling

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ABSTRACT

This paper introduces BlendCon, a fully automated framework capable of simultaneously synthesizing and labeling construction imagery data. This framework simulates a construction site by orchestrating 3D mobile objects against a 3D background and produces multimodal labels for target entities. The effectiveness of the synthetic data in training object detection models was thoroughly validated. For the construction worker detection task, a YOLOv7 model trained with synthetic data nearly matched the performance of a model trained with real data: it achieved 71% AP@0.5–0.95 compared to 75% for the real data-trained model. Moreover, the model trained with synthetic data surpassed its real data counterpart in scenarios requiring stricter IoU thresholds, particularly above 85%. Acquiring a sufficient quantity and diverse range of imagery data has been a primary challenge in construction studies that focus on automation and digitization through deep neural networks. BlendCon can significantly contribute to addressing this data scarcity challenge.

1. Background and motivation: unlocking the potential of visual AI in construction

The construction industry, while still in the early stages of transformation, is embracing visual AI solutions. This can be marked by a growing number of studies exploring potential applications of Deep Neural Network (DNN)-based computer vision such as progress monitoring [1–3], safety monitoring [4], quality control [5], and the cutting-edge concept of live digital twinning [6]. This trend of visual AI's wide adoption in construction is propelled by three major factors. Since 2016, there has been a notable evolution in DNN architectures and the development of more varied DNN training algorithms [7]. Concurrently, computing hardware, such as GPUs, has continued to increase in power while becoming more accessible and affordable [8], with cloud-based solutions offering affordable computation [9]. Additionally, more diverse benchmark datasets with new modalities tailored for emerging vision tasks have become increasingly available online. Datasets like ImageNet [10] with more than 1200 thousand, COCO [11] with more than 330 thousand or Open Image Dataset [12] with more than 15 million labelled images have set a great development platform in the computer vision society.

Despite these advancements, the pace of research into visual AI for

the construction sector lags behind other major sectors, creating an ever-widening gap. While the computer science domain continually introduces new DNN architectures, novel training algorithms, innovative training platforms, and new data modalities, adoption and implementation in construction research have been slow. One instance of this lag is the limited availability of public datasets. For example, we found only two limited public datasets specifically for object detection in construction: Moving Objects in Construction Sites (MOCS) [13]; and the Large-Scale Small Object Detection Dataset (SODA) [14] with only 40 and 20 thousands datapoints respectively. This study attempts to address the primary reason behind this slow progress: the absence of a sufficient number and diverse range of DNN training data specifically crafted for construction-centric applications.

2. The bottleneck: how data scarcity affects DNN excellence

At the core of visual AI lies supervised DNN models, which are replete with millions of learnable parameters [15]. Success in training these models relies heavily on the quantity and diversity of training data, and this scarcity of DNN training data has been a significant hurdle for many construction studies. Typically, the volume and diversity of training data used in previous construction studies were insufficient to

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fully saturate a DNN architecture, leading to invalid conclusions [13,14]. Expecting a DNN model trained on such a limited dataset to achieve performance comparable to one trained on a multimillion-image dataset is, from a theoretical standpoint, untenable.

Given that every construction project is both unique and ever-changing, developing effective DNN models requires training with large-scale datasets that capture a wide spectrum of backgrounds, objects, materials, equipment, and workers at various scales, viewpoints, and illumination conditions [16]. Despite its critical importance to DNN optimization, assembling a sufficiently large set of construction scene training images has so far proved logistically impossible. Manual data collection and labeling are prohibitively costly and time-consuming (e.g., segmentation of a single image on Google AI Platform costs approximately \$2.72 [17]). Also, there are cases where labeling data necessitates deploying physical sensors (e.g., motion capture sensors for 3D pose labeling), which is not feasible on real construction sites. Last but not least, the sharing of construction site images is not allowed in many cases due to issues of confidentiality.

Although many construction studies have invested significant resources into manual data collection and labeling, these datasets are typically small and biased. These low-quality datasets result in overfitted DNN models of low accuracy and scalability [18], miss opportunities to develop deeper DNN architectures [19], and do not allow for competitive benchmarking against other research outcomes due to the lack of standardized training, validation, and test datasets [18]. Recognizing these challenges, the potential of synthetic datasets as a solution to data scarcity in construction is increasingly recognized.

3. Research question: can synthetic data be the answer to the data scarcity problem?

Unlike human vision, DNNs can effectively interpret synthetic images that visually characterize realistic scenes, making them a viable alternative to real images for training purposes [20]. This perspective opens a novel approach to preparing DNN training images—the use of computational image synthetization and automated labeling. The holy grail of this approach is to automatically generate and label non-real but real-looking construction scene images using a process that does not require any manual inputs or site visits. This can create large datasets for DNN training, saving time and resources. Currently, this innovative approach to addressing the serious gap in construction training data remains underexplored.

This technique capitalizes on computational power to synthesize new data from analogous sources. One of the most well-known studies in this domain, known as the ‘Flying Chairs’ research [21], generated synthetic images using 3D models of chairs in a virtual world, proving effective in boosting DNN training. The process includes automatic and simultaneous labeling of each image, allowing for virtually unlimited datasets [22], as exemplified by the SYNTHIA dataset [23]. Another recent benchmark study validated the effectiveness of synthetic data on DNN training by using the engine of the ‘GTA’ video game to generate a large dataset for multi-person human detection and tracking [24]. This experiment demonstrated that well-crafted synthetic data can effectively replace real data for vision tasks such as pedestrian detection, re-identification, segmentation, and tracking. This growing body of research supports the proposition that synthetic data, provided it is sufficiently realistic, can serve as a viable substitute for real-world images in training DNNs [24].

Despite its potential, synthetic image generation and labeling for DNN training in construction is still in early developmental stages. Previous attempts using building information modeling (BIM) for data generation have often omitted key elements, such as human workers, and faced challenges with multimodal labeling and the realism of synthetic images. This study aims to bridge these gaps.

The following outlines the three specific research gaps that were identified and addressed in our research:

- **Worker-centric dynamic simulation:** Field workers, the key players of a construction project, are the key elements of productivity and thereby the major target to monitor. The most critical research gap that the authors identified at the outset of this study was the lack of a computational tool that can simulate virtual construction workers with realistic motions. Recent studies, such as those by Acharya [25], Ma et al. [26], Hong et al. [27], and Ying et al. [28], leveraged virtual data that building information models generate. However, those studies were required to scope down to static structural components as the platforms used did not allow for modeling of the dynamic mobility inherent in construction resources. Recent endeavors by Soltani et al. [29], Kim and Kim [30], and Mahmood et al. [31], have used 3D modeling tools to synthesize virtual construction equipment data from varied viewpoints. While these efforts addressed construction equipment, the dynamic presence of field workers with realistic construction motions remains largely unexplored. Only a few studies, for example, Neuhausen et al. [22], have explored the inclusion of human workers, but in limited settings involving limited diversity in background textures, workers' motions, and avatar designs, combined with confined imaging conditions. The previous literature highlights this evident research gap and calls attention to the need for a reproducible method to simulate and label realistic construction workers in motions [22].
- **Multimodal labeling:** The integration of diverse sensory data, such as coupling depth maps with RGB images [32] or merging semantic segmentation masks with other imagery data [33], has been recognized as a promising approach for enriching DNN training processes [34]. Despite this, construction studies have encountered challenges in capitalizing on these advancements, primarily due to a dearth of suitable multimodal datasets. For multimodal data to effectively augment DNN training, all modalities must be collected at the same viewpoint, not to mention that an extensive amount of data is required [35]. Manually compiling such datasets presents significant hurdles, particularly due to the dynamic nature of construction environments resulting in viewpoint or timeframe inconsistencies in collected data. This challenge highlights the need for innovative solutions to multimodal labeling under dynamic imaging conditions.
- **The reality gap:** The level of reality (or fidelity) of synthetically generated images is paramount in DNN training. The “reality gap” is a term that refers to the divergence between the distributions of real and synthetic images as a result of their disparate origins [36]. Earlier construction research that attempted to use synthetic images in DNN training overlaid 3D synthetic avatars onto a 2D real background image [37,38], which often compromised the fidelity of the resulting outcome. This approach faced challenges in convincingly placing avatars within 2D backgrounds, a task that can be more suitably addressed in a 3D environment (refer to Fig. 1). Additionally, the resulting discrepancy in lighting conditions between the avatar and the background was evident. Avatars rendered under unique lighting conditions contrasted with their backgrounds, inadvertently providing DNNs with misleading cues during training—a visual cue absent during real-world inference. When present, such discrepancies ultimately diminish the effectiveness of DNN models trained with synthetic images in real-world applications, underscoring the need for high fidelity approaches to synthetic image generation.

4. Research objectives

This study had two main objectives: (i) developing an end-to-end automated framework to synthesize and label multimodal construction images, and (ii) conducting DNN training experiments to validate the effectiveness of the synthetic data. We first developed the computational framework from scratch, incorporating extensive source coding, and named it BlendConstruction (BlendCon), which offers several novel functionalities.

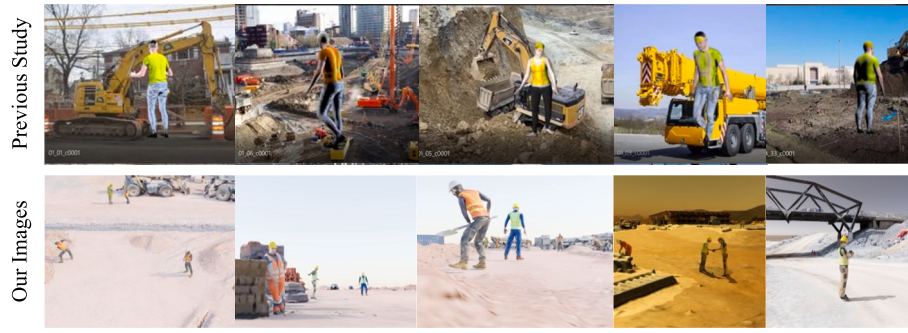


Fig. 1. Samples of synthetic construction images: prior study [37] vs. ours.

- Worker-centric simulation: BlendCon integrates dynamic elements, simulating construction workers with realistic motions under varying camera viewpoints, thus maximizing dataset diversity. While BlendCon is capable of simulating other construction equipment along workers, the focus in this study was on worker-centric simulation and validation due to its complexity and research gap.
- Multimodal labeling: BlendCon generates various labels, including 2D and 3D bounding boxes, 2D/3D poses, semantic segmentation masks, and depth maps. It automatically generates precise labels simultaneously with image generation.
- Mitigated reality gap: BlendCon enhances image quality by incorporating 3D backgrounds and simulations, ensuring realistic avatar placements (e.g., on the floor, not floating in air; see Fig. 1) and harmonized lighting conditions, thus providing high-fidelity outcomes.

Following the development of BlendCon, we conducted DNN training experiments to test the effectiveness of the synthetic data in terms of trainability and scalability. For trainability, through a qualitative verification, we confirmed that DNNs could be trained from scratch with synthetic data at levels comparable to real data through

visual verification. For scalability, we quantitatively evaluated the performance of DNNs trained on synthetic data in real-world scenarios by comparing it to models trained on real data using various quantitative metrics.

The body of this paper provides details on the technical aspects of BlendCon (Section 5 below) and presents the validation procedure and the results achieved as measured by the trainability and the scalability of the synthetic data-trained model (Section 6). Finally, our key insights (Section 7) and conclusion (Section 8) are presented.

5. BlendCon: the construction-centric worker-focused computational framework for simultaneous image synthetization and multimodal labeling

BlendCon is built upon the Blender graphic simulation engine [39] (Fig. 2), with its operation structured in two primary stages: (i) image synthetization; and (ii) multimodal labeling.

- Stage #1 - image synthetization: BlendCon initiates a synthetization process, loading a meticulously designed realistic horizon (e.g., a sky model; Fig. 2). Subsequently, a pre-processed construction scene (e.g.,

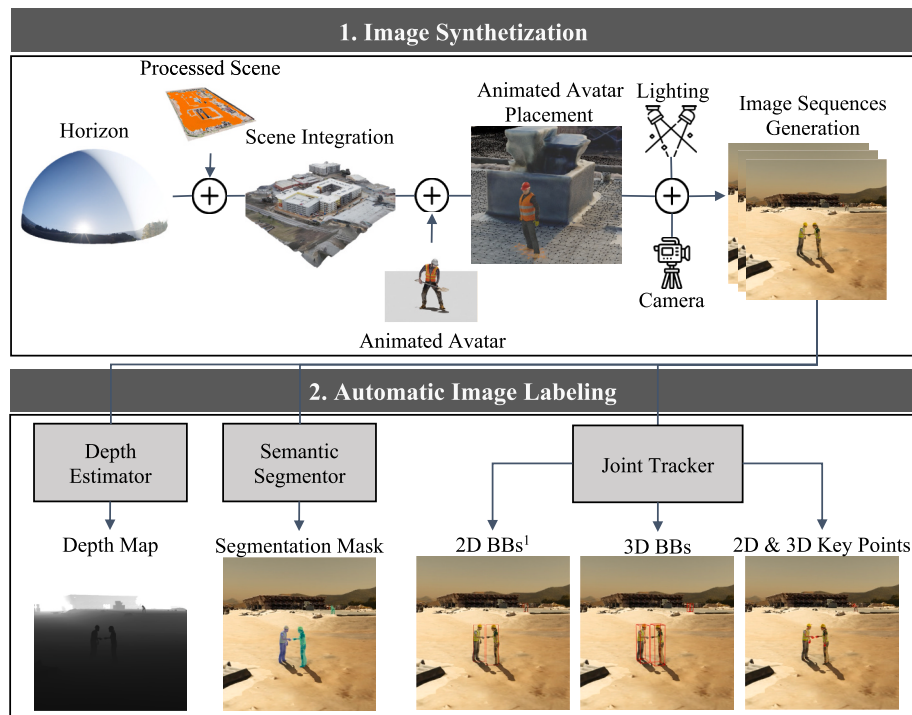


Fig. 2. Overview of BlendCon.

Note: BB(s) stands for bounding box(es).

g., a 3D point cloud), which is randomly chosen from the back-end data archive, is encapsulated inside of the horizon. Following this, a designated number of animated avatars with motions are positioned within predefined locations within the scene. The camera parameters, including camera distance, focal length, and resolution, along with lighting variables, such as time of day, atmospheric conditions, and the sun's angle, are then meticulously configured. Following these configurations, BlendCon sets and simulates a virtual construction scene, which is in turn used to render synthetic image sequences, generating the first major research outcome.

- **Stage #2 - multimodal labeling:** Having a computational and mathematical simulation as the base, BlendCon retrieves and tracks precise details of all elements and hyper-parameter values activated during the simulation. This retrieved data serves as the foundation for subsequent mathematical processing that generates multimodal labels. This stage manages the conversion of raw scene element data (e.g., camera's extrinsic and intrinsic parameters and kinematic information of animated avatars) into a series of ground truths.

The following describes the specific stages BlendCon uses to synthesize and label construction scene images.

5.1. Stage #0: preparation of Back-end data archive

Prior to the two main stages, there is Stage #0, which involves archiving back-end data (e.g., pre-processed 3D point clouds and animated avatars). BlendCon completes simulations by blending these data; therefore, the diversity of simulations BlendCon can produce is directly influenced by the diversity of the back-end data. This subsection offers potential users a set of guidelines for preparing the necessary back-end data, and details the extent and nature of data the authors prepared for this study.

The primary inputs to BlendCon are animated 3D construction worker avatars (foreground) and 3D construction scenes (background). These inputs require preprocessing to facilitate the full automation of BlendCon. Fig. 3 illustrates an overview of the data preparation process, which is described in detail in the rest of this subsection.

5.1.1. Preparation of 3D construction avatars animated with motions

We followed a three-step procedure as illustrated in Fig. 3(a).

- **Step #1, collecting 3D static avatars:** We collected pre-designed static avatars from online sources [40–42]. These avatars are meticulously designed by expert artists, yet static without any motions embodied. To boost the diversity, we expanded our selection to varied body shapes, heights, weights, and the range of skin tones, as illustrated in Fig. 4. In this study, we collected 37 avatars: 18 avatars resemble construction workers and the remaining 19 avatars wear non-construction clothing. We included this mix not only to mirror the typical attire found on construction sites but also to add richness to the diversity of our archived data.
- **Step #2, collecting construction motion data:** Motion is fundamental to spatio-temporal simulations of mobile objects such as construction



Fig. 4. Sample of avatars archived.

workers. To collect our own motion capture (MoCap) data, we utilized a wearable motion capture (MoCap) system (Smartsuit Pro, Rokoko [43]), specifically emphasizing actions commonly observed at construction sites (Table 1). We collected 15 types of construction MoCap data, each resembling typical construction activities, including bricklaying, material handling and carrying, shoveling, hammering, drilling, cutting, and more. Each MoCap data is collected at 100 frames per second (FPS).

- **Step #3, avatar-motion mapping:** This process involves retargeting MoCap data to static avatars using a specialized algorithm. Built on Rokoko's retargeting function [44], this method matches each bone location of the avatars with the corresponding location in the motion data. A series of normalizations are applied to both the motion data and avatars, ensuring a compatible naming convention between them. Once normalized, each joint movement in the avatar is determined by its corresponding joint in the motion data for each frame. As a result, we successfully archived a total of 555 animated construction worker avatars (37 static avatars $\times$ 15 MoCap data = 555 animated avatars).

5.1.2. Preparation of 3D construction scenes

In BlendCon, a 3D scene serves as a digital background where

Table 1

List of all animation collected and used for this study.

Motion Type	Duration (unit = seconds)
Brick Laying	19.5
Carrying Wheelbarrow	13.7
Checking Design Sheet	16.1
Curing Concrete	17.5
Cutting Plate	15.9
Digging	20
Drilling	16.5
Hammering – Standing	17.7
Hammering – Bending Knees	18.1
Hammering – Knees on Ground	17.2
Loading Wheelbarrow	9.4
Looking Around	5.9
Shaking Hand	1.9
Walking	15.5
Leaning on Wall	27.2

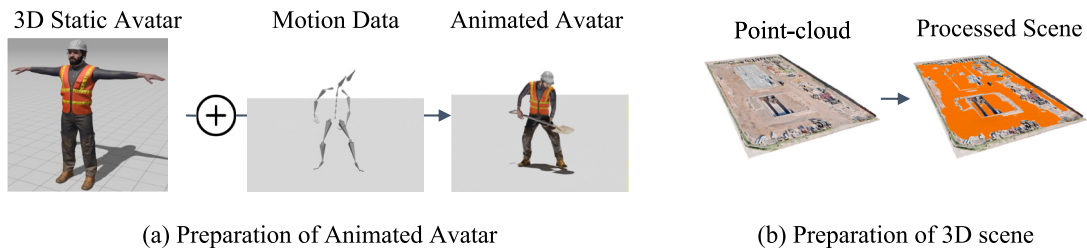


Fig. 3. Overview of BlendCon's input data pre-processing.

animated avatars are deployed. This background can simply be a colored and meshed 3D point cloud; however, to facilitate the initial stages of the simulation process it needs to be pre-processed. First, these digital arenas need to be scaled to real-world dimensions. Second, a 'work zone' needs to be defined within each 3D point cloud to determine plausible areas where animated avatars can stand up and navigate through (e.g., the orange shaded areas in Fig. 5). We collected 47 instances of 3D point clouds from online source [40–42] each of which captures a unique construction site. Then, we manually adjusted the scale and defined work zones for each 3D scene. The pre-processing ensures that all 3D point clouds in our set are accurately aligned and scaled, realistically, within the fully automated BlendCon. When pre-processing of all these instances was completed we had realized 647,971 m² of work zone. Fig. 5 visualizes some samples of the scene dataset and their corresponding work zones, highlighted in orange.

5.2. Stage #1: image synthetization

Image synthetization, the first main stage of BlendCon, consists of four major steps: (i) setting up an arena; (ii) positioning a camera; (iii) setting up lighting conditions; and (iv) rendering.

5.2.1. Step #1: setting up an arena

The data synthetization process begins with setting up a realistic arena. First, a real-looking horizon is added to an empty space (Fig. 6), noting that this horizon functions as a boundary. A pre-designed horizon is used, consisting of a vast half-sphere with a high-dynamic-range image mapped onto it. The half-sphere provides a visual representation of the atmosphere and horizon simultaneously. Next, a processed scene is picked randomly from our back-end data archive (Fig. 6). This scene already matches to the real-world scene scale and comes with a pre-defined work zone, setting the simulation arena for the animated construction avatars. Subsequently, a user-specified number of avatars are randomly selected from the back-end data archive and placed within the predefined work zone (Fig. 6). BlendCon allows users to determine the number of avatars to be placed in a simulation.

5.2.2. Step #2: positioning the camera

Once global scene setup is complete a camera is placed in the scene (Fig. 6). The camera's primary role is to project 3D points on the global coordinate system back to a local 2D image plane (i.e., pixel coordinate system) through extrinsic and intrinsic transformations. This section explains how a camera is set up before rendering, with an emphasis on defining the appropriate extrinsic and intrinsic camera parameters, as these settings are essential to achieving a high degree of realism.

To ensure the presence of at least one worker in the camera's view, a virtual camera is set up around a primary avatar who is chosen at random. Functionally, this is achieved by positioning a camera on a sphere with a radius specified by a user, centered on the target avatar (see Fig. 7). Our heuristic experiments and observations led us to define a realistic range for the radius of this virtual sphere. We selected the radius from a normal distribution, with an average of 6 m and a variance of 18 m.

BlendCon also allows users to define intrinsic camera parameters, namely its focal length and resolution. Through heuristic experiments, we determined that a focal length spanning from 32 to 70 mm yields realistic images suitable for our research. The choice of resolution is largely influenced by computational capacity. The larger the image, the more computation needs to render a single image. In our study, we opted for an image resolution of 1920 by 1920 pixels. To render this resolution of synthetic image in an RTX 3080 GPU it takes around 1 min per image. In summary, Table 2 details the camera parameters used.

As a final step we added an occlusion measurement function to camera positioning in BlendCon. This is essential to ensure a clear or partially unobstructed view of the target avatar. By calculating the occlusion percentage of the target avatar from the camera's perspective, this function helps BlendCon place the camera with an optimal view of the scene, ensuring the target avatar remains unblocked by any scene meshes. This function is also beneficial for users who intentionally want to simulate occlusion data, although it was not addressed in this study.

5.2.3. Step #3: setting the lighting conditions

Lastly, lighting conditions need to be set up before the final rendering of the camera-ready simulation (Fig. 6). To this end, we employed a sky rendering model proposed by Nishita et al. [45],

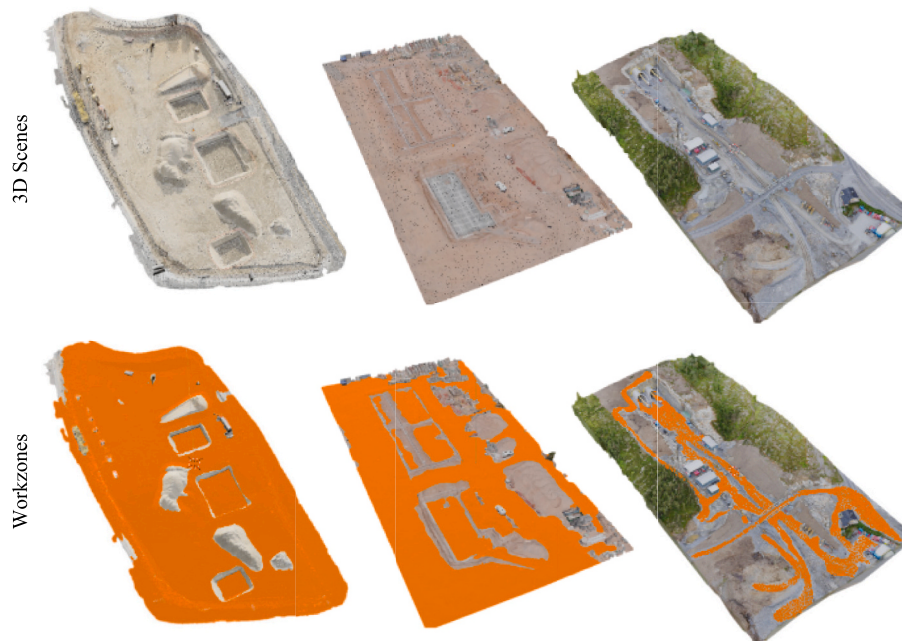


Fig. 5. Samples of 3D scenes with their corresponding 'work zones' designed for BlendCon.

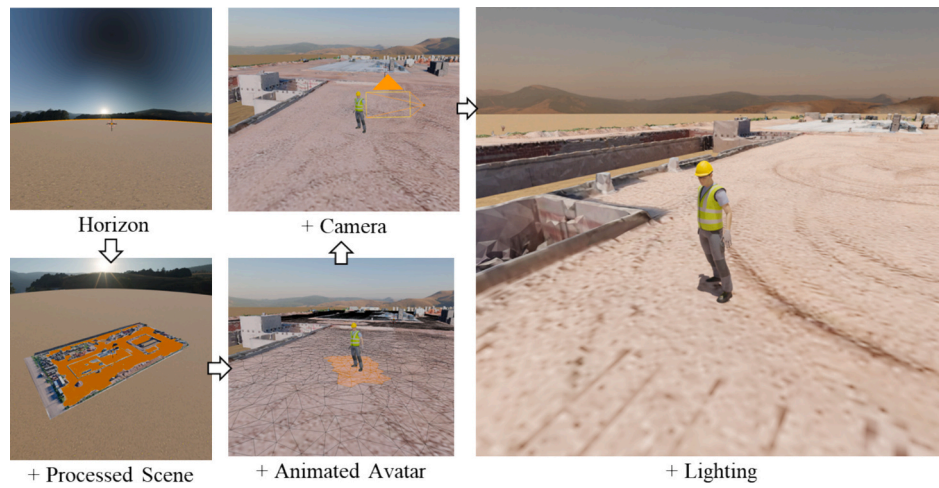


Fig. 6. Procedural set up of arena.

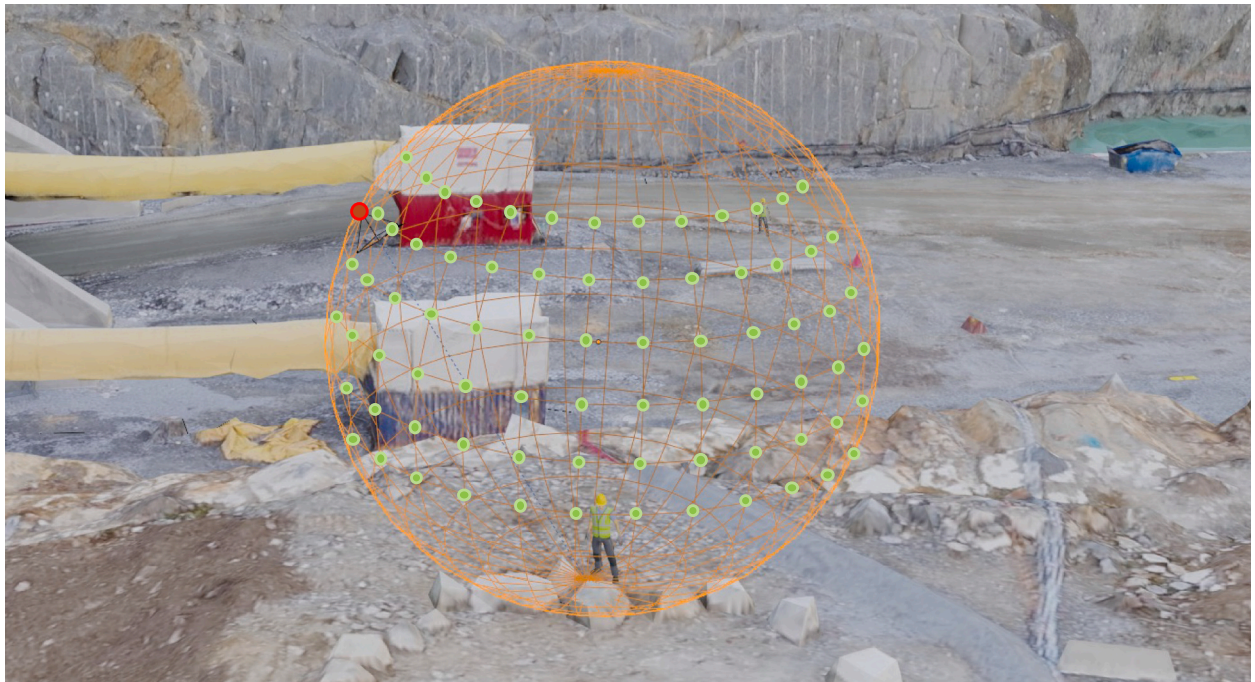


Fig. 7. Camera positioning on a user-defined sphere around an animated avatar in BlendCon.

Note: On the sphere, green circles show some of the available positions for camera locations, while red circle shows the randomly selected camera location.

Table 2

Overview of default setting of BlendCon for camera parameters.

Camera Parameter	Type	Values (Default; User-adjustable)
Location	Extrinsic	Localized around target avatar with distance with $N(\mu = 6, \sigma^2 = 18)$
Orientation	Extrinsic	Set to track the head of the primary target avatar
Focal Length	Intrinsic	Uniform range between 32 and 70 mm
Resolution	Intrinsic	1920 by 1920

Note: N represents a normal (Gaussian) distribution.

implemented in Blender as a 'sky texture' [46]. This model facilitates the simulation of sky lighting with several parameters including sun size, intensity, elevation, rotation, altitude, density of air molecules, dust and water droplets, and ozone molecules (Table 3). The Nishita model is

Table 3

Overview of lighting-related parameters in BlendCon.

Lighting Parameter	Value
Sun Size	0.545
Sun Intensity	0.1
Sun Elevation	$\frac{\pi}{2} \times N(\mu = 0.5, \sigma^2 = 0.16)$
Altitude	0
Density of Air Molecules	$10 \times W(\alpha = 1, \beta = 1.1)$
Density of Dust and Water Droplets	Clear day atmosphere
Density Of Ozone Molecules	Clear day atmosphere

Note: N represents a Normal (Gaussian), and W represents a Weibull distribution.

particularly advantageous in this case because it provides a physically accurate representation of atmospheric scattering, resulting in highly realistic sky colors. By adjusting the parameters of this model, BlendCon is able to render different lighting scenarios, from clear skies to overcast conditions, and simulate the full spectrum of natural lighting conditions throughout the day. This flexibility ensures that our simulations can account for various environmental conditions, enhancing the realism and accuracy of our renderings.

In this research we aimed for photo-realism. To this end, we utilized the default values for sun size, altitude, dust density, water droplet density, and ozone molecule density as suggested by Nishita et al. [45]. For the remainder of the parameters, we used values determined through our heuristic experiments. A sun intensity of 0.1 best suited our day-time realistic simulation. Sun elevation, which represents the sun's angular displacement from the horizon in degrees, was observed to provide realistic daytime lighting when values were sourced from a Gaussian distribution with a mean of 0.5 and a variance of 0.16. The last hyperparameter was the density of air molecules which is critical in capturing the reddish hues of the sky at dawn and dusk [45]. For this study, the ideal density was determined experimentally by multiplying ten with a value sourced from a Weibull distribution, which had a shape parameter of one and a scale parameter of 1.1.

5.2.4. Step #4: rendering

Once the arena (i.e., scene stage), players (i.e., animated construction worker avatars), a camera, and imaging conditions have been set up, BlendCon starts synthesizing images. Rendering of a scene is enabled by a render engine called Cycles [47]. This rendering engine uses NVIDIA-Optix [48] to construct images from the camera's viewpoint. Each image pixel is constructed by sampling ray traces. These rays propagate within the scene, reflecting off materials until they either encounter a light source or hit a set bounce limit [49]. The resulting light capture is translated into pixel values in the image. BlendCon uses an adaptive sampling algorithm [47] to automatically sample ray traces of lights for each pixel to produce images with a controlled level of noise threshold. As a hyperparameter, we set the noise threshold to 0.01 in this study. The noise threshold can be any value between 1 and 0.0001. Tweaking this threshold can speed up the rendering process at the expense of image quality, or conversely, enhance image quality at the cost of rendering time.

Another key hyperparameter is the frame sampling rate. Images are generated in sequences, drawing from the animations of the avatars. Users can determine this sampling rate. In our study we selected a rate of 10 frames per second. With the motion capture data used in BlendCon captured at 100 frames per second (FPS), our research extracted one frame for every 10 frames of the animation sequence.

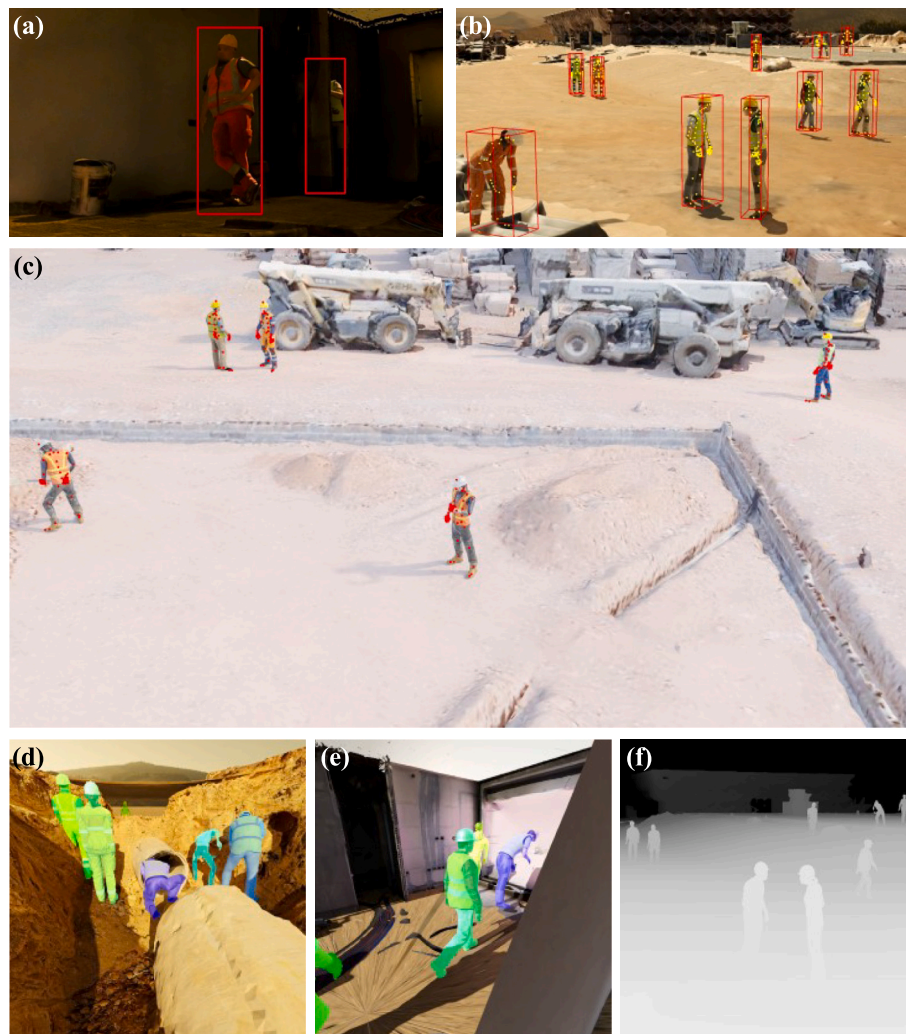


Fig. 8. Illustration of BlendCon's multimodal annotation capabilities.

Note: Visualization of (a) 2D bounding boxes, (b) 3D bounding boxes and keypoints, (c) keypoints, (d and e) segmentation masks, and (f) depth map translating distance information into grayscale values.

5.3. Stage #2: multimodal labeling

Multimodal labeling in computer vision refers to the process of annotating images with multiple types of information or labels that come from different modalities. These annotations can include spatial coordinates of body joints, known as 3D [50] and 2D [51] keypoints, which are necessary to understand body positions and movements. Similarly, object detection relies on 3D [32,52,53] and 2D [54] bounding boxes to define the parameters of objects in three-dimensional and/or two-dimensional space. For tasks that necessitate depth perception, depth maps [55] provide the distance between objects and the camera. And finally, semantic segmentation masks [56] are utilized to uniquely identify and classify each object in a scene, enhancing the model's ability to understand complex visual inputs. Fig. 8 illustrates the different types of labels generated by BlendCon, showcasing how each modality provides distinct yet complementary information for comprehensive scene understanding.

To ensure accuracy and consistency in multimodal labeling, BlendCon employs three distinct coordinate systems. First, there is Blender's global coordinate system, the natural reference frame for scenes or objects within Blender. Second, the camera's coordinate system originates on the image plane, with the Z-axis extending outwards, perpendicular to this plane. This system represents the camera's viewpoint, with world coordinates transformed to camera view coordinates through rotation and translation defined by the camera's extrinsic parameters in Blender. Lastly, the pixel coordinate system projects the 3D world into 2D, as perceived by the camera, with coordinates in pixels and the origin at the top left corner of the image.

Crucially, the labels generated by BlendCon are both calibrated and synchronized. Calibration ensures that spatial information is consistently matched across different modalities, providing an accurate representation of objects in the scene. Synchronization guarantees that these modalities are aligned in time, maintaining temporal consistency and ensuring that dynamic events are accurately captured and represented.

Table 4 offers a comprehensive overview of the labels supported by the current version of BlendCon. This table not only delineates each label type but also specifies their corresponding coordinate systems and the functions responsible for generating these labels.

The labeling process of BlendCon consists of three distinct functions: the (i) joint tracking function; (ii) depth estimation function; and (iii) semantic segmentation function. These functions play an essential role in the accurate depiction of the scene, and the rest of this subsection will delve into the technical details of each function.

5.3.1. Joint tracking function

The joint tracking function of BlendCon is designed to capture the kinematic details of 3D animated worker avatars by extracting precise 2D and 3D labels from their bone structures, which are analogous to human skeletal systems. This process is depicted in Fig. 9.

The joint tracking function begins with tracking the joint position of

avatars that are already placed in work zones of the scene. This function gets the 3D coordinates of the avatars' joints in Blender's global coordinate system (see Fig. 9 (a)). The function then translates these 3D coordinates from Blender's global coordinate system into the camera's coordinate system (see Fig. 9 (b)). This step generates the labels needed to train 3D pose estimation models. These coordinates are expressed as x, y, and z values in the camera's coordinate system.

Next, the function employs these coordinates to outline the 3D bounding boxes around the workers. To determine the precise dimensions of the 3D bounding boxes, we calculated the extremities of the keypoints in the camera's coordinate frame (see Fig. 9 (c)), which is essential when predicting 3D bounding boxes in 3D coordinates [53].

To meet the requirements of DNN models requiring pixel-level prediction [32,52], the joint tracking function also transforms these 3D bounding boxes defined in the camera coordinate system into 2D projections within the pixel coordinate system. This is achieved by computing a transformation matrix from the camera's intrinsic and extrinsic parameters, projecting the 3D information onto a 2D pixel grid (see Fig. 9 (d)).

The function then proceeds to extract 2D keypoints, projecting the 3D joint locations onto the pixel coordinates, resulting in a set of 2D coordinates that serve as training labels for models focused on 2D human pose estimation (see Fig. 9 (e)). Subsequently, it applies a min-max selection algorithm to these 2D keypoints to deduce the 2D bounding boxes (see Fig. 9 (f)).

5.3.2. Depth estimation function

The depth estimation function in BlendCon translates the distances of scene objects from the camera lens into grayscale pixel values, a crucial step for creating depth maps. This transformation is accomplished using Blender's 'map range node,' which acts as a converter, mapping the distance of an object from the camera to a pixel value on a scale from 0 to 255. This scale corresponds to the grayscale spectrum, allowing for the visualization of distance through shades of gray. Our empirical tests have shown that setting the maximum input range to 100 m results in the most accurate depth maps, as it optimizes the representation of distances up to that point. Distances beyond 100 m are assigned the maximum grayscale value, appearing as white in the depth map. By applying this mapping technique, BlendCon can render detailed depth maps that accurately reflect the spatial layout of a scene. It should be highlighted that because the depth map is processed using the same viewpoint as the corresponding RGB image they can be used as a set, i.e., as a 4-channel training source for both 2D and 3D computer vision tasks.

5.3.3. Semantic segmentation function

Semantic segmentation is a process in computer vision where each pixel in an image is classified into distinct categories, delineating different objects and elements within the scene. As currently realized within BlendCon, it specifically involves classifying each pixel as part of an individual avatar, the sky, or the background (i.e., any object except avatars and sky in the scene), thus distinctly separating the avatars from other elements. To achieve this, the semantic segmentation function capitalizes on Blender's object index rendering feature. This feature enabled us to assign distinct indices to each avatar individually, while the sky and the background were indexed separately. To assign indices, we used the names of objects to distinguish between them. We set the 3D scene to have index of 50 and the horizon to be zero. Then, for each avatar, we assign a value starting from 254, decremental by steps of 10 for each avatar. These indices are then converted into pixel values within the grayscale spectrum. The resulting image, composed of these object-specific indices, enables the precise segmentation of each object based on their unique pixel values.

5.4. BlendCon: a scalable solution for diverse use cases

The simplicity and flexibility inherent in BlendCon's design

Table 4
Comprehensive summary of BlendCon's labeling capabilities.

Label Type	Coordinate System	Function Involved	Data Type
3D Keypoint	Camera Coordinates	Joint Tracking	[x, y, z]
2D Keypoint	Pixel Coordinates	Joint Tracking	[x, y]
3D Bounding Box	Camera Coordinates	Joint Tracking	[x, y, z]
3D Bounding Box	Pixel Coordinates	Joint Tracking	[x, y]
2D Bounding Box	Pixel Coordinates	Joint Tracking	[x, y]
Depth Map	Pixel Coordinates	Depth Estimator	Grayscale Image
Semantic Segmentation	Pixel Coordinates	Semantic Segmentation	Grayscale Image

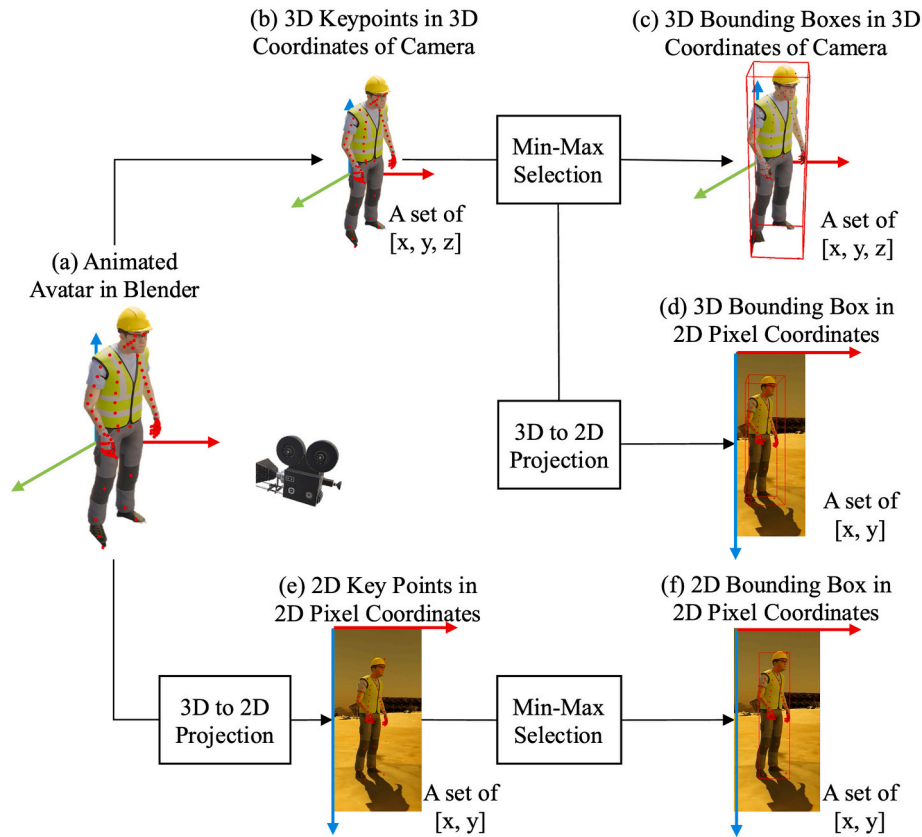


Fig. 9. Joint tracking function's data pipeline.

Note: Just like real humans, each worker in BlendCon is designed with a bone structure. The joint tracking function leverages these bones and joints to extract labels.

contributes significantly to its scalability, enabling it to effectively meet the demands of diverse use cases, whether in research or industrial settings. To confirm the usability of BlendCon in an academic context, we installed and ran the framework to generate a small dataset comprising 25,000 images. For this purpose, we utilized three NVIDIA GPUs (RTX3080s) on our local machines, and the framework was successful in generating the required dataset for our specific use case. The framework demonstrated similar success in an industrial context at a considerably larger scale. Our industry partner, utilized BlendCon to generate 985,660 images using 260 virtual machines on Amazon AWS service within 2 days. This successful large-scale application was a significant milestone in confirming both the usability and flexibility of BlendCon in the hands of third-party users, as well as the ability of the platform to successfully scale.

6. Validation of the effectiveness of synthetic construction data: trainability and scalability

To assess the impact of synthetic construction data on DNN training we conducted two experiments, with one focusing on trainability and the other scalability. Below is an outline of our experimental design:

- **Assigned computer vision task and selected architecture:** 2D object detection is central to modern computer vision pipelines. In addition to being a core function, 2D object detection serves as a precursor for advanced tasks like 3D object detection and 2D/3D pose estimation, or action recognition. Our validation experiments targeted the 2D detection of construction workers, employing the YOLOv7 architecture [54]. Proven by previous researches [57–62], YOLOv7 is particularly favored among construction researchers due to its accessibility and immediate applicability, alongside its accuracy and computational efficiency.
- **Application scenario:** While designing the experiments, we decided to focus on a practical scenario where an autonomous ground vehicle (AGV) requires 2D worker detection DNN. Basically, we intended to develop 2D worker detection DNNs for the AGV using two distinct datasets: one consisting solely of synthetic data, and the other using exclusively real data. This approach was intended to demonstrate the genuine value of synthetic data in the development of DNNs for real-world commercial applications. For the purposes of this research, the objective of the trained DNNs was set to reliably detect construction workers within a 20-m range of the AGV.
- **Training approach:** To achieve a pure comparison between synthetic and real data, we meticulously trained the YOLOv7 models from scratch across all experiments. In addition, our heuristic observations led us to set our training duration to 800 epochs, which would typically be considered quite long. We noticed that terminating the training at earlier epochs often resulted in the most recent model being the best, suggesting that the saturation point had not been reached. Therefore, to ensure comprehensive learning and to accurately gauge the model's potential we extended the training to 800 epochs. This ensured the models could fully assimilate the dataset, crucial for an unadulterated evaluation of synthetic versus real data training outcomes. Moreover, we decided to keep the image size on 1024 and 960 when training yolo models with synthetic or real images, respectively.
- **Evaluation metric:** We narrowed our focus to the Average Precision (AP), which has been widely employed in computer vision studies. The AP combines the meaning of both precision and recall, offsetting potential bias that individual metrics could yield. It computes the average of precision values for varying recall levels across multiple thresholds. This comprehensive measure reflects the accuracy and completeness of the detections, making it suitable for evaluating scenarios demanding high recall (detecting as many objects as

possible) and high precision (ensuring these detections are accurate). While adopting the AP, another consideration was the extent of Intersection over Union (IoU). While AP at the IoU of 0.5 (AP@0.5) is a common benchmark — balancing leniency and strictness — it does not suffice, especially in evaluating industry-grade AIs. In our experiments, we chose to assess APs from 0.5 to 0.95 IoUs to capture a nuanced understanding of the model's performance across various localization precision requirements. AP@0.5–0.95 (i.e., the average of the APs across the IoU of 0.5–0.95) was also adopted for a single-shot comparison.

6.1. Synthetic data generation and real data collection

Creating or obtaining suitable datasets is a fundamental prerequisite for any machine learning experiment. Our experiments used BlendCon to generate the synthetic dataset, and our team assembled the real dataset through careful curation and combining two public benchmark datasets.

- **Generating the synthetic dataset:** Using BlendCon with the hyperparameter settings described previously in this paper, we rendered 1578 sets of construction simulations, resulting in a synthetic dataset with 24,008 images which in turn contained 40,322 bounding boxes. Fig. 10 provides samples of the generated synthetic data.

- **Assembling the real dataset:** We utilized two public benchmark datasets as our starting point: (i) MOCS [13]; and (ii) SODA [14]. Both datasets are large-scale and tailored for 2D object detection. We selectively filtered these datasets, retaining images featuring human workers along with their corresponding labels, as dictated by our focus on worker detection. By combining images from MOCS and SODA we prepared a diversified and balanced real-image dataset, containing a total of 22,773 images with 74,026 bounding boxes. This integrative approach was deliberately chosen to mitigate potential biases that could arise from relying exclusively on a single dataset. To keep the consistency, we converted all the real images into 960×960 by adding gray boundaries whenever needed.
- **Filtering process:** Both synthetic and real datasets underwent a series of filters equally. First, filters were applied based on bounding box sizes to suit our use case: worker detection within 20 m. Validation and test subsets excluded bounding boxes smaller than 9% of the image size, while training subsets removed those smaller than 4%. This approach was based on heuristic observations: workers within 20 m typically have bounding boxes over 172 pixels, about 9% of a 1920×1920 image. Extending training sets to include images with bounding boxes down to 4% of image size was found to enhance model learning. Exclusively for real dataset, additional filters were applied to sort out night-time shots, blurred, cropped, watermarked images, and aerial drone shots.
- **Dataset splitting:** In splitting the synthetic dataset between the training, validation, and test sets, we took precautions to prevent any data leakage caused by scene contexts. The data was split based on

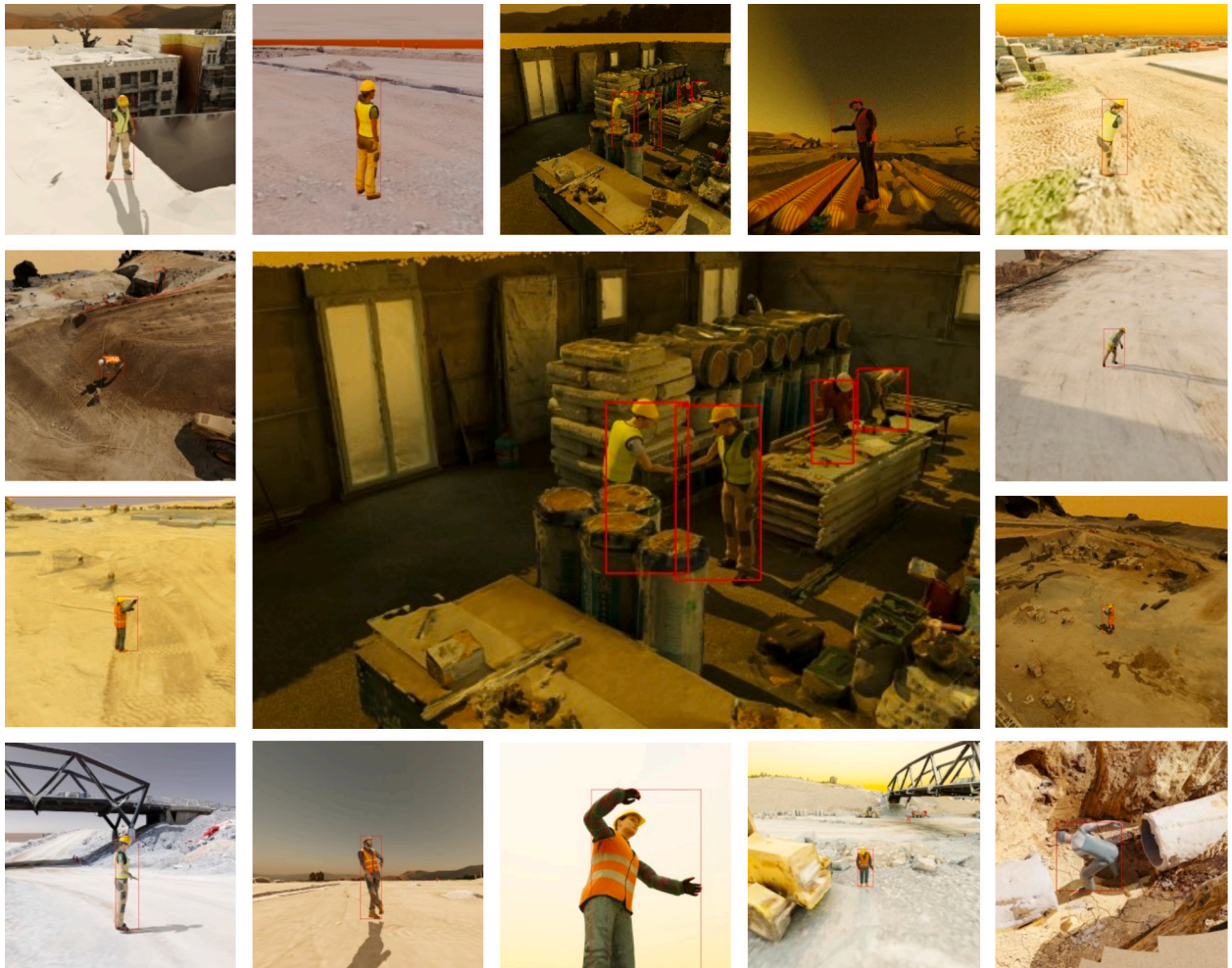


Fig. 10. Samples of synthetic images with 2D bounding boxes

video names, not individual images, ensuring that during training, the model would not be exposed to familiar scene contexts present in the test or validation sets. As a result, the synthetic training set had 35,708 bounding boxes; the validation set contained 3689, and the test set had 2917 bounding boxes. We also split the real dataset based on bounding box counts, sampling images for the test and validation sets until we reached a near thousand bounding boxes for each. The remaining pool of images constituted the training set.

The number of images and bounding boxes in each of the synthetic and real datasets is provided in Table 5.

The distribution of bounding boxes in the real and synthetic datasets is demonstrated in Fig. 11. In this figure, each bounding box is represented as a single dot. The horizontal axis illustrates the ratio of the bounding box width to the image width, while the vertical axis shows the ratio of the bounding box height to the image height. A small empty square at the bottom right of the graph is evident due to the data filtering process. Additionally, kernel density estimates are plotted on the top and right marginal axes, which demonstrate a close proximity in the peak of the bounding boxes in both datasets.

6.2. Validation of trainability

A YOLOv7 model was trained with and validated on synthetic data. For comparison, another model was trained with and validated on real data. Our goal was to visually examine the training loss history for both models to identify any similarities or differences in training patterns between synthetic and real data. Fig. 12 shows training and validation losses for both instances. YOLO's loss function comprises box loss, object loss, and class loss [54]. Class loss is a measure of how accurately the model can identify the correct category of an object in multi-class detection tasks. However, given our dataset's focus on a single class, class loss becomes irrelevant in our context. Box loss measures the accuracy in predicting target object bounding boxes, while object loss evaluates the model's ability in identifying target object presence within these boxes. The total loss linearly combines box, object, and class losses [54].

Fig. 12 provides a comparative analysis of the learning curves for both the synthetic and real YOLOv7 models, which was central to our 'trainability' validation objective. It demonstrates that the synthetic model's learning trajectory mirrors that of the real model, with both exhibiting similar patterns in their loss reduction over epochs.

When examining box loss, both models display a rapid decline in the initial phases of training, indicative of quick adaptation and efficient bounding box prediction. This convergence suggests that both datasets equip the model with sufficient information for identifying object boundaries early in the training process.

In terms of object loss, the early plateau in the synthetic model's learning curve suggests it may not encounter the same dataset complexity as the real data. Here, 'complexity' pertains to the dataset's variety and intricacy. This diversity is critical, as it pressures the model to generalize better to real-world diversity. A less complex dataset may lead to quicker mastery of simpler patterns, as indicated by a steep initial decline in box loss for the synthetic model. However, this could hinder the model's performance in more varied real-world scenarios. Despite these concerns, the parallel trends in object loss for both datasets indicate that the synthetic dataset's simplicity does not significantly impede

the model's learning of object detection, suggesting that synthetic data remains a viable training resource.

Furthermore, a consistent decrease in total loss during the early training phase is visible for both models, reinforcing the notion that synthetic data can be as effective as real data for initial learning stages. These observations affirm that training with synthetic data is not significantly different from training with real data in terms of the overall training process and loss patterns. This similarity bolsters our confidence in the efficacy of synthetic data for training DNN models, as it appears to provide a comparable learning experience to real-world data.

6.3. Validation of scalability

To set the stage for our scalability experiment, we crafted a multi-tiered benchmarking approach. The objective was to analyze the impact of training data volume on model performance and to establish where the synthetic model stands compared to real models trained on datasets of varying size. We segmented the real data pool into smaller, incremental datasets, training multiple YOLOv7 models to create a spectrum ranging from data-scarce to data-rich conditions. We achieved this by sampling a series of datasets, each with different sizes for the training set, but with identical validation and test sets. These training sets were sampled from the pool, ranging from 5000 to 35,000 samples, in increments of 5000. This partitioning strategy, detailed in Table 6, enabled us to not only scrutinize the performance gradient across differently trained models but also to pinpoint the synthetic model's relative position in terms of data sufficiency and model robustness.

To provide a fair basis for comparison, we set a limit on our largest real dataset of 35,000 bounding boxes. This decision was taken to prevent the real dataset from surpassing the synthetic dataset. With our seven distinct real datasets at hand, we proceeded to train seven different YOLOv7 models. These models were all trained from scratch, using YOLOv7's default hyperparameters. The AP@0.5–0.95 of these models on the real test dataset are displayed in Fig. 13. The result obtained from the synthetic model is displayed as a dashed line. It should be noted that the DNN model trained with synthetic data was also validated with synthetic data, with no real data of any kind used during the development stage, ensuring that the results observed were achieved solely from synthetic data.

Upon analyzing the AP@0.5–0.95, a notable observation emerges. The models trained solely on synthetic data, despite their lack of exposure to real-world scenarios, outperform those trained with limited real data. Further, when the same amount of training samples used (i.e., around 35,000 bounding boxes in this case), the AP@0.5–0.95 difference between the synthetic and real models was merely about 4%. The promising performance of the synthetic model, achieved without any exposure to real images, highlights its potential. It demonstrates the synthetic model's ability to effectively bridge the reality gap and adeptly generalize to the unfamiliar domain of the real world.

Another critical observation is the direct correlation between the volume of training data and the performance of real models. This trend solidifies the idea that the efficacy of DNN models is deeply intertwined with the abundance of data. In scenarios of data scarcity, even the most potent models may falter. It was evident that performance of the models was sub-optimum and stagnant when using training datasets containing 5000 or 10,000 bounding boxes. However, increasing the size of datasets beyond this threshold realized a significant surge in model efficiency, suggesting a pivotal point in the learning curve.

Upon examination, the performance of a synthetic model trained on a dataset of 35,708 labelled bounding boxes was comparable to a real model trained on a dataset of 20,000 bounding boxes. As the amount of training data increases beyond 20,000 images, real models improve but not dramatically so. This gap in the performance of the synthetic and real DNN models is presumably a manifestation of the reality gap, and it does not significantly widen even with enhanced training data for real models. We posit that this slight disparity can be bridged either by

Table 5

Number of samples in synthetic and real datasets.

Dataset Name	Data Type	Training	Validation	Test
Synthetic	Bounding Box	35,708	3689	925
	Image	20,997	2336	675
Real	Bounding Box	74,026	998	925
	Image	19,862	418	391

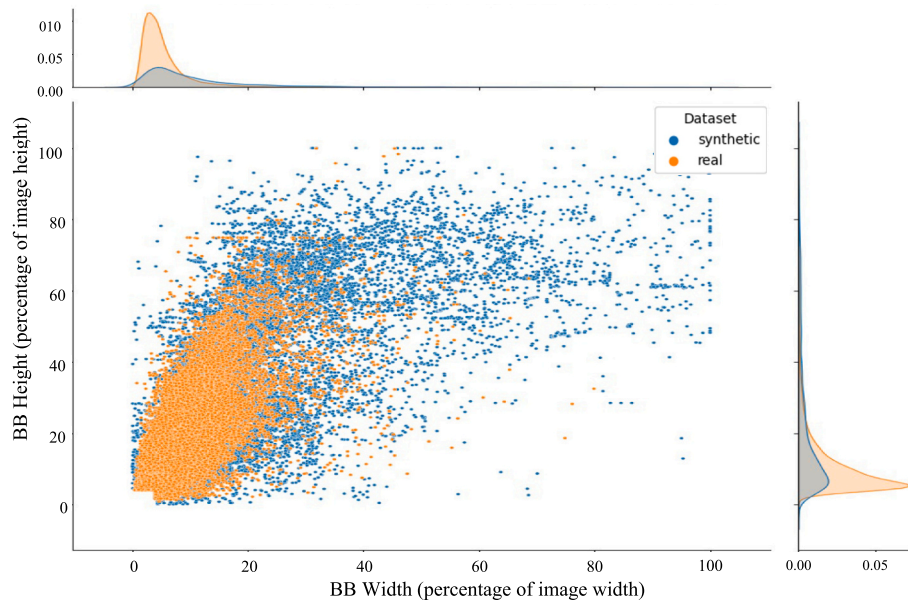


Fig. 11. Distribution of bounding box size ratio in real and synthetic datasets.

Note: Each bounding box is represented as a single dot. Also, kernel density estimates are plotted on the top and right marginal axes.

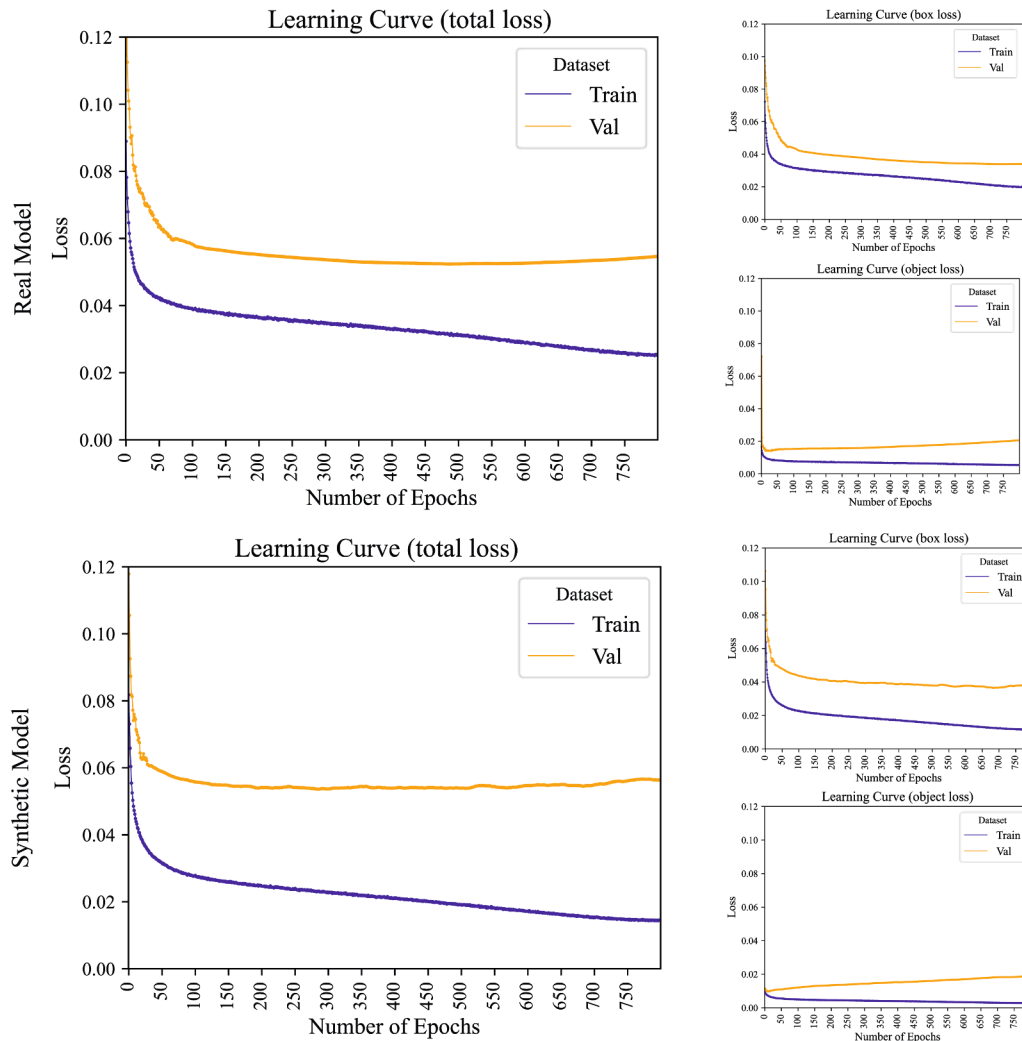


Fig. 12. Learning curves of real and synthetic models.

Table 6

Number of bounding boxes in each training, validation, and test datasets.

Dataset Name	Training	Validation	Test
Synthetic	35,708 (synthetic)	3689 (synthetic)	925 (real)
Real 5 K	5004	998	925
Real 10 K	10,001	998	925
Real 15 K	15,007	998	925
Real 20 K	20,005	998	925
Real 25 K	25,003	998	925
Real 30 K	30,002	998	925
Real 35 K	35,017	998	925

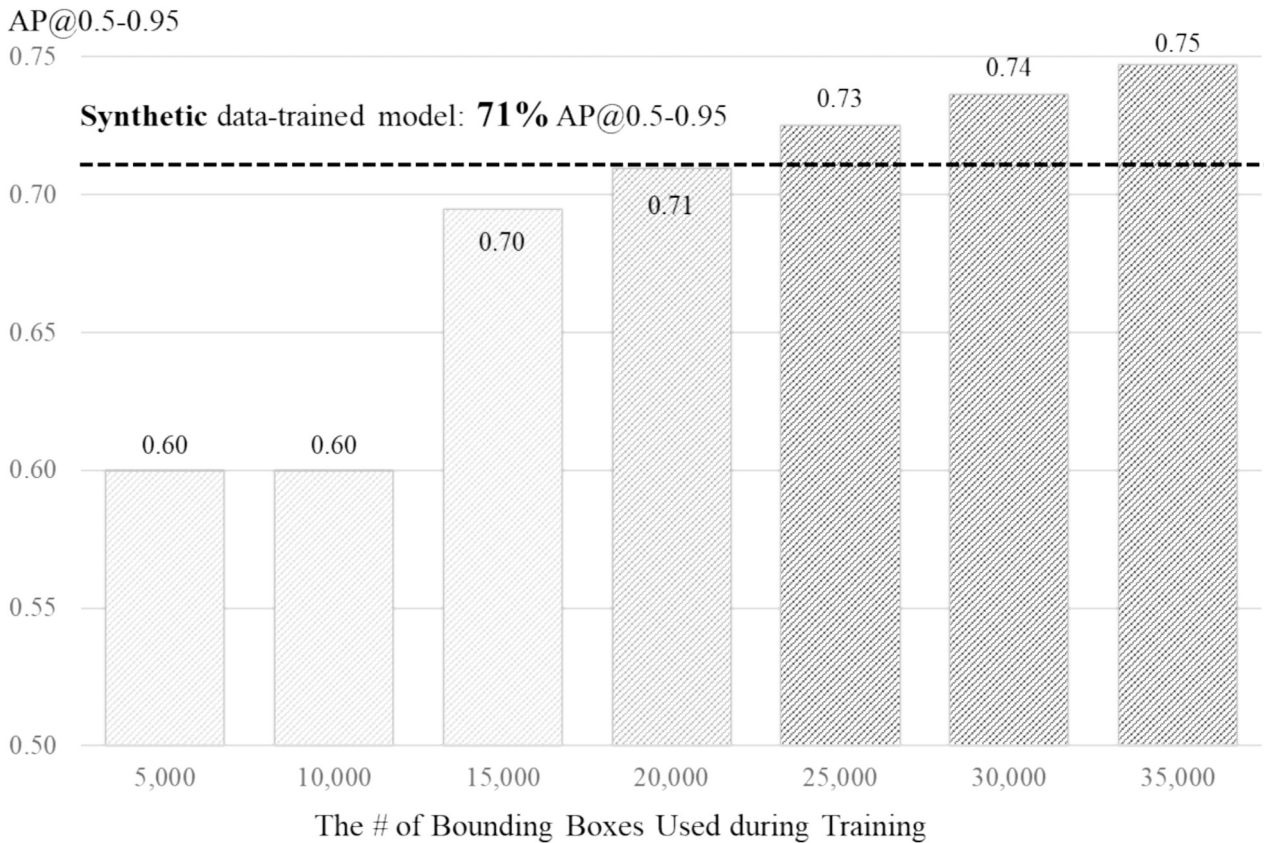
Note: It was assumed that no real data are available while training the synthetic model. Thus, no real data were used while training the synthetic model, even for validation. Also, note that each data point represents a bounding box, and due to multiple bounding boxes per image, the total may slightly exceed the originally targeted numbers.

augmenting the diversity or the volume of synthetic data, or by implementing strategies to reconcile domain differences, transitioning smoothly from synthetic to real scenarios.

To interpret these results with more detail, we evaluated the AP at each IoU threshold value, from 0.5 to 0.95. Table 7 presents the APs of all models.

Table 7 sets the stage for a deeper analysis by comparing the precision of detection at various degrees of overlap between predicted and ground-truth bounding boxes. Fig. 14 delineates the AP scores over a range of IoUs, plotting the synthetic model's performance against that of the real models.

Based on Fig. 14, the synthetic model initially appears to underperform compared to the real models, particularly at lower Intersection over Union (IoU) thresholds. This trend persists until an IoU of 0.6, where the synthetic model reaches the performance level of the real model trained with the smallest data volume. As the IoU threshold increases, the synthetic model shows progressive improvement, surpassing all real models at an IoU of 0.85. This pattern may be attributed to the inherently precise bounding boxes of synthetic data. In contrast, real-world data labeling, often prone to human error, may result in less accurate bounding box definitions. This becomes more noticeable at higher IoU thresholds, where precision is critical. The decline in AP for

**Fig. 13.** AP@0.5–0.95 of real and synthetic models on real test dataset.**Table 7**

Average precision scores at different IoU thresholds.

Model Name	IoU Thresholds									
	0.5	0.55	0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95
Synthetic	0.88	0.86	0.84	0.81	0.79	0.76	0.71	0.63	0.52	0.27
Real 5 K	0.91	0.89	0.86	0.82	0.76	0.67	0.56	0.36	0.16	0.02
Real 10 K	0.91	0.89	0.86	0.81	0.75	0.66	0.55	0.36	0.16	0.01
Real 15 K	0.95	0.94	0.92	0.89	0.84	0.79	0.69	0.55	0.32	0.06
Real 20 K	0.96	0.95	0.92	0.89	0.85	0.79	0.71	0.57	0.36	0.09
Real 25 K	0.97	0.96	0.93	0.91	0.88	0.82	0.71	0.58	0.38	0.12
Real 30 K	0.97	0.96	0.94	0.91	0.88	0.80	0.73	0.60	0.41	0.16
Real 35 K	0.97	0.96	0.95	0.92	0.89	0.83	0.74	0.61	0.44	0.16

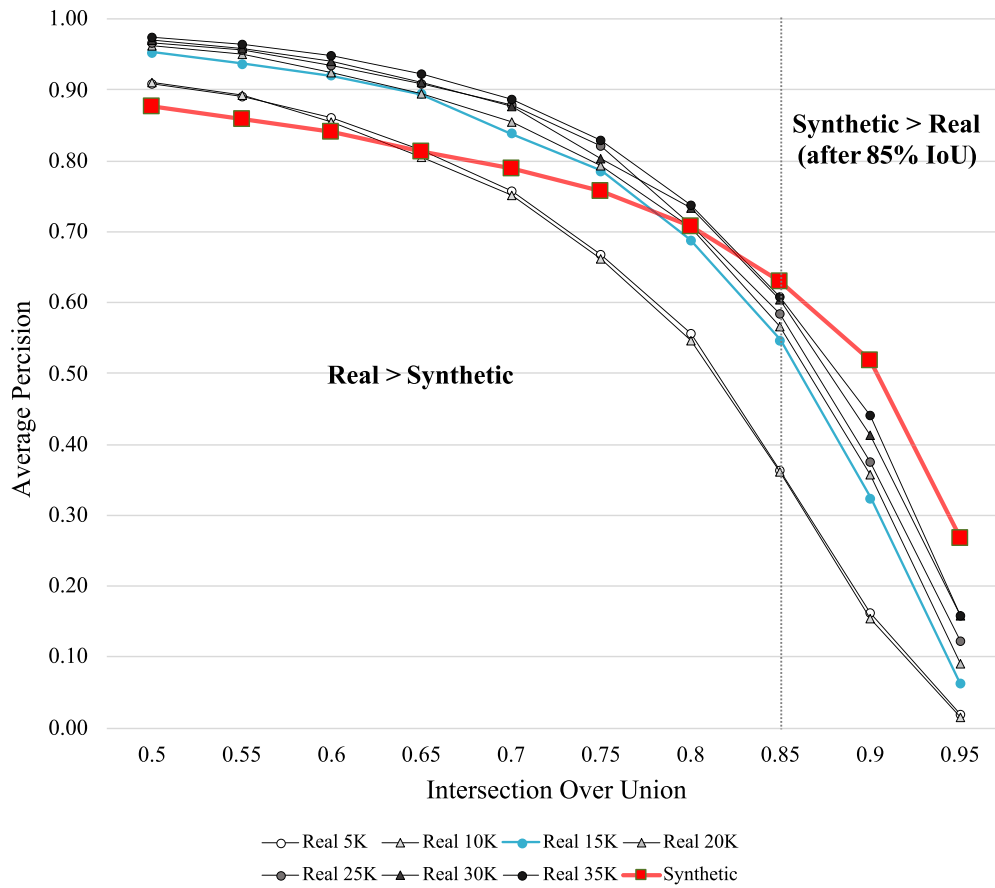


Fig. 14. Synthetic vs. Real Model Performance.

real models is steeper than for the synthetic model as IoU thresholds rise, underscoring the value of the meticulous and precise labeling that synthetic data offers. It leads to more consistent detection results, even with tightening IoU thresholds. This observation suggests that for tasks requiring high localization accuracy, investing in the quality of data labeling could be as important, if not more so, than merely increasing the quantity of training data—a potentially significant insight for future research.

To visually examine the performance of models trained on real and synthetic data, we selected examples of both successful and erroneous predictions. Fig. 15 shows instances where the synthetic model

outperforms the real model, correctly identifying workers and handling occlusions better.

Conversely, Fig. 16 highlights scenarios where the real model excels, particularly with occluded workers and varied clothing. The synthetic model struggles with color distortions and non-standard worker appearances.

It should be noted that these images serve as illustrative examples of the strengths and weaknesses of the two models. But more extensive experiments are needed to draw definitive conclusions.

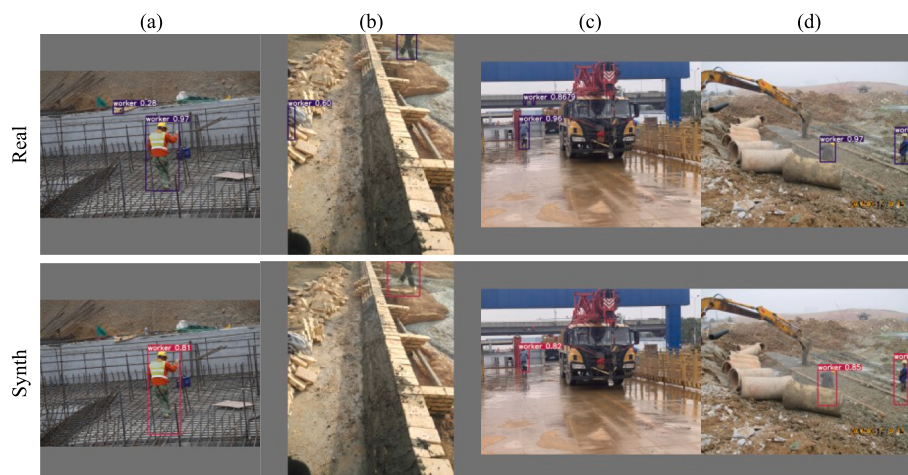


Fig. 15. Cases where the synthetic model outperforms the real model.

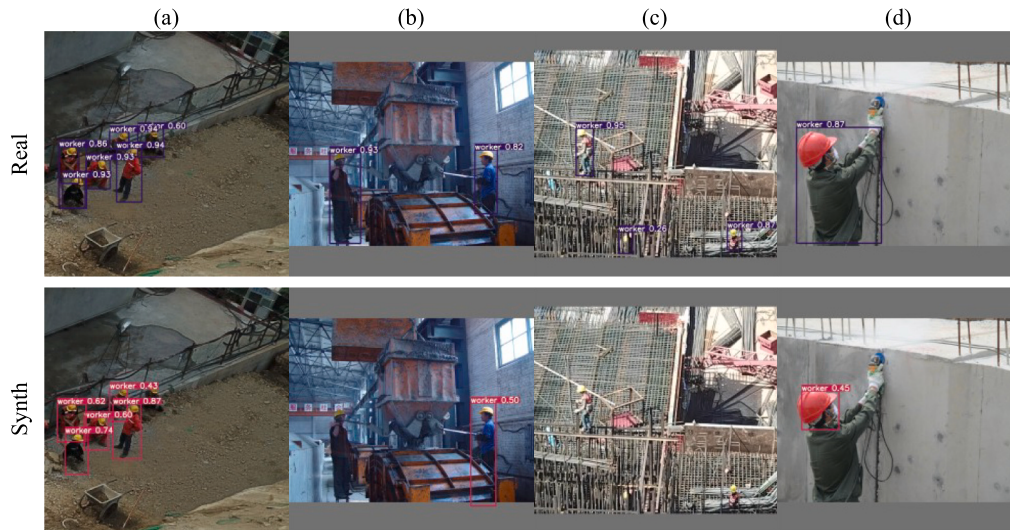


Fig. 16. Cases where the real model outperforms the synthetic model.

7. Discussion: the impact of synthetic data on construction-centric DNN

This section explores the potential implications and impacts of using synthetic data in DNN training within the context of the construction industry. Based on the results of our study, we discuss the significant opportunities that the use of BlendCon opens up for researchers and industry professionals. We also highlight a number of unexpected findings, new challenges, and future research directions.

7.1. Unleashing potential: the implications of BlendCon in DNN training

Our study unearthed a important observation. Synthetic data holds the potential to serve as a primary dataset in DNN training. The parity in performance between models trained on synthetic data and their real-data counterparts, especially when assessed using average precision, is a compelling testament. Such findings underscore the capability of BlendCon-generated synthetic images to effectively replace real images. This positions BlendCon as a uniquely valuable tool in the toolkit of DNN training methodologies.

Moreover, synthetic data proved to be an economical alternative to traditional data collection methods. Our experiments suggest that achieving the DNN performance synthetic data provides would require over 20,000 real images. Synthetic data's ease of generation—merely a few clicks away, particularly with cloud computing—stands in stark contrast to the hefty costs and efforts of collecting and labeling real data. For example, just the manual labeling process for segmentation masks costs around \$2.27 per image, costing well over \$40,000 for a 20 K dataset. Meanwhile, synthetic data, being readily available at the click of a button, eliminates such financial and logistical burdens.

Additionally, employing synthetic data enhances the efficiency of training DNNs for rare but significant scenarios. Tailored synthetic data can be created for unique DNN use-cases, bypassing the challenges of data collection and privacy concerns associated with real construction sites while facilitating training for uncommon but crucial situations. For instance, there might be limited real-world data on specific safety incidents like proximal human-robot collaboration on sites. Sole reliance on this sparse real data isn't viable. Synthetic data, however, offers cost-effective simulation of such situations, allowing for the development of DNNs adept at managing even highly uncommon and unlikely scenarios.

7.2. Looking ahead: the future with BlendCon

In addition to BlendCon's contribution to the training of DNN models

for visual AI, this tool opens up new avenues of academic research. Extrapolating from our study's outcomes, we see vast new horizons for synthetic data-led development by both academic researchers and industry professionals. The following spotlights a series of unanticipated insights, emergent challenges, and prospective trajectories in harnessing synthetic data for DNN training endeavors.

7.2.1. BIM-integrated BlendCon: a journey into the future to generate prospective images

One of BlendCon's standout features is its adaptability to diverse input data. BlendCon needs a 3D scene as an arena for the animated avatars. This scene sets the background of the generated images. In this study, we used point clouds as the primary source of the 3D scene. However, the construction sector offers another valuable resource: BIM. BIM files are fully compatible and can be utilized as inputs for BlendCon's 3D scene creation. Tapping into BIM enables the generation of synthetic images with backgrounds that mirror the actual structure to be constructed. Given that building information models are conceived and realized during the early phases of a building's lifecycle, integrating them with BlendCon essentially offers a glimpse into the future, generating imagery from sites yet to be constructed. This forward-thinking image generation allows DNN models to familiarize themselves with forthcoming construction landscapes. In essence, by integrating BIM with BlendCon, we may be able to equip DNN models with a temporal advantage, enabling them to gain acquaintance with future sites.

7.2.2. Beyond single modalities: embracing multimodal data generated by BlendCon

An increasing number of studies indicate that DNN models trained with multimodal data demonstrate improved performance [35]. Multimodal DNNs leverage diverse data inputs—such as RGB, depth, and segmentation indexes—to create more comprehensive data representations, thereby enhancing their capability to perform complex visual tasks. However, a significant hurdle is the dearth of multimodal datasets, which stems from the intricacies involved in both the spatial calibration of data across various sensors and the temporal synchronization of data streams [35]. BlendCon addresses these challenges by efficiently generating and aligning multimodal data for individual scenes, streamlining the process of data collection and labeling. This capability broadens the spectrum of vision tasks that can be addressed. For instance, by integrating multiple modalities into a single training sample—like combining RGB with depth maps for improved 3D perception, or depth with segmentation for advanced spatial recognition—BlendCon paves the way for DNNs to tackle a wider array of applications, from

night vision enhancement to complex spatial analyses.

7.2.3. Active learning with BlendCon: advancement of DNN training paradigms

Beyond its image generation capabilities, BlendCon's engine offers adaptability. While this study focused on the applicability of the data generated by the platform, it's also worth noting that the flexibility of the platform itself is a valuable asset. It is a stepping stone towards integrating the platform more directly with the DNN training frameworks. This integration could open the possibility of automated, continuous, and iterative learning for DNN models. We could design a training framework wherein a DNN model identifies its failure cases, maps these failures to the inputs of BlendCon, and generates synthetic images corresponding to that failure. The model could then enrich the existing dataset with synthetic data that is specifically tailored to the observed failure case and retrain the model for continuous improvement. This potential for automated learning and adaptation could revolutionize the way we approach DNN training and significantly enhance model performance over time.

7.3. Investigation of BlendCon's potential enhancements

When considering how BlendCon's functionality can be enhanced, three areas stand out: (1) diversity; (2) level of reality; and (3) in-depth characterization and validation of synthetic data.

In synthetic data generation for DNN training, diversity is critical [20]. This encompasses the diversity of avatar poses and shapes, clothing, motions, and background scenes. The more diverse the input to the data generation process, the more inclusive the features in the resulting data, leading to better generalization in real-world applications. However, providing this level of diversity in the data generation process is a challenge. The repository of open-source 3D assets suitable for data generation is not only limited but often comes entangled with licensing complexities that can impede the DNN training workflow. To truly harness the potential of BlendCon in enhancing DNN training, a deliberate investment in developing a high-quality dataset of diversified 3D assets is essential. This involves the accumulation of detailed 3D backgrounds and the creation of dynamic, lifelike animated avatars that closely mimic the appearance and actions of actual construction workers. Our efforts to expand this library of back-end data will be progressive and ongoing.

Realism is the other significant area for improvement. Some studies suggest that the realism of synthetic data plays a vital role in DNN training [63]. The argument posits that increasing the realism of generated images can help DNN models bridge the "reality gap" between synthetic and real domains when deployed in real-world applications. To address the reality gap, recent efforts include advanced generative models that introduce targeted noise into synthetic images to mimic real-world imperfections. With the emergence of controllable diffusion models, these techniques have become more practical, showing promise in increasing the realism of synthetic datasets. However, the challenge of realism extends beyond just the fine details and noise patterns of the images. Enhancing realism would require complex modeling of real-world dynamics, which can be particularly challenging. To enhance realism, the interactions of avatars with their environment need to be modeled in detail. This would include simulating complex scenarios, where multiple entities interact with one another. It would also require the adaptation of worker motions based on their specific working environments. Last but not least, modeling more realistic clothing would be another challenge. In our study, we used fixed meshes attached to the avatars as clothes. To simulate the real world more accurately, we would need to adopt a new approach that allows the modeling of realistic clothing for construction worker avatars.

Our validation experiment with BlendCon was just the beginning. Many more validation experiments are necessary to understand the relationship between input data distribution and DNN performance.

Given that a major contribution of BlendCon is the automated multimodal labeling, DNN training experiments for other labels and vision tasks will follow. The next stage of validation will include 3D object detection, 2D and 3D pose estimations, semantic segmentation, and depth estimation. We also plan to tackle more challenging vision tasks, such as detecting extremely small-sized workers from dynamic aerial viewpoints and conducting 4-channel (i.e., RGB + D) training to enhance both daylight and potentially night vision capabilities. Throughout these validation experiments, we will perform more in-depth analysis, such as characterizing the relationship between synthetic imaging conditions (e.g., viewpoint, illumination, scale of foreground) and the DNN's final inference performance. Last but not least, there is untapped potential for improving DNN model performance by combining synthetic and real data. Previous studies [26] demonstrate that this combination can be more effective. Our efforts to understand and identify the genuine potential of synthetic data will continue.

With these advancements in follow-up studies, BlendCon will have another chance to improve its effectiveness.

8. Conclusion

The construction industry, faced with enduring challenges in productivity and safety, stands to benefit greatly from advancements in DNN-based visual AI. However, the industry's lack of access to high-quality, diversified data significantly limits this potential. This study addresses the roadblock by presenting BlendCon, a fully automated computational framework designed to generate synthetic data for DNN training. By harnessing the power of graphic engines, BlendCon can create a realistic virtual replica of a construction site, generating non-real yet realistic images. The framework's design allows for full control over image properties, lighting conditions, and avatar customizations, enabling a high degree of diversity and flexibility in the synthetic data generated.

Through this research, we not only introduce a computational (synthetic) data generation solution that eliminates the time-consuming manual data collection process, but also demonstrates the potential of this synthetic data in training scalable, field-applicable DNNs. Our experimental findings validate the trainability of models using synthetic datasets and demonstrate that synthetic data can indeed act as a viable substitute for real data, specifically in the task of construction worker detection. The research team will continue to improve BlendCon, aiming to enhance its diversity and realism. Future DNN training experiments across diverse 2D and 3D computer vision tasks will also be conducted to fully understand the relationship between synthetic data and DNN performance.

With the usability and validity of BlendCon established, the research team identifies several promising future research directions. Integrating BlendCon with BIM could facilitate project-specific data generation, while coupling it with a generative AI could further diversify the synthetic data produced. As BlendCon supports depth labeling, it could also be utilized for multimodal DNN training (e.g., RGB + D) for 3D computer vision tasks. Additionally, the closed-loop active DNN training introduced in the discussion (Section 7) presents a bold yet promising area for further investigation. There could be more research opportunities to be uncovered. The BlendCon codebase will be made publicly available in the project's GitHub repository: <https://github.com/Al-Tohidifar/BlendCon>.

CRedit authorship contribution statement

Ali Tohidifar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Daeho Kim:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **SangHyun Lee:**

Writing – review & editing, Supervision, Formal analysis, Conceptualization.

Declaration of generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work the authors used ChatGPT4.0 in order to spot and revise grammatical errors and typos in our draft manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

Daeho Kim reports financial support was provided by Natural Sciences and Engineering Research Council of Canada. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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